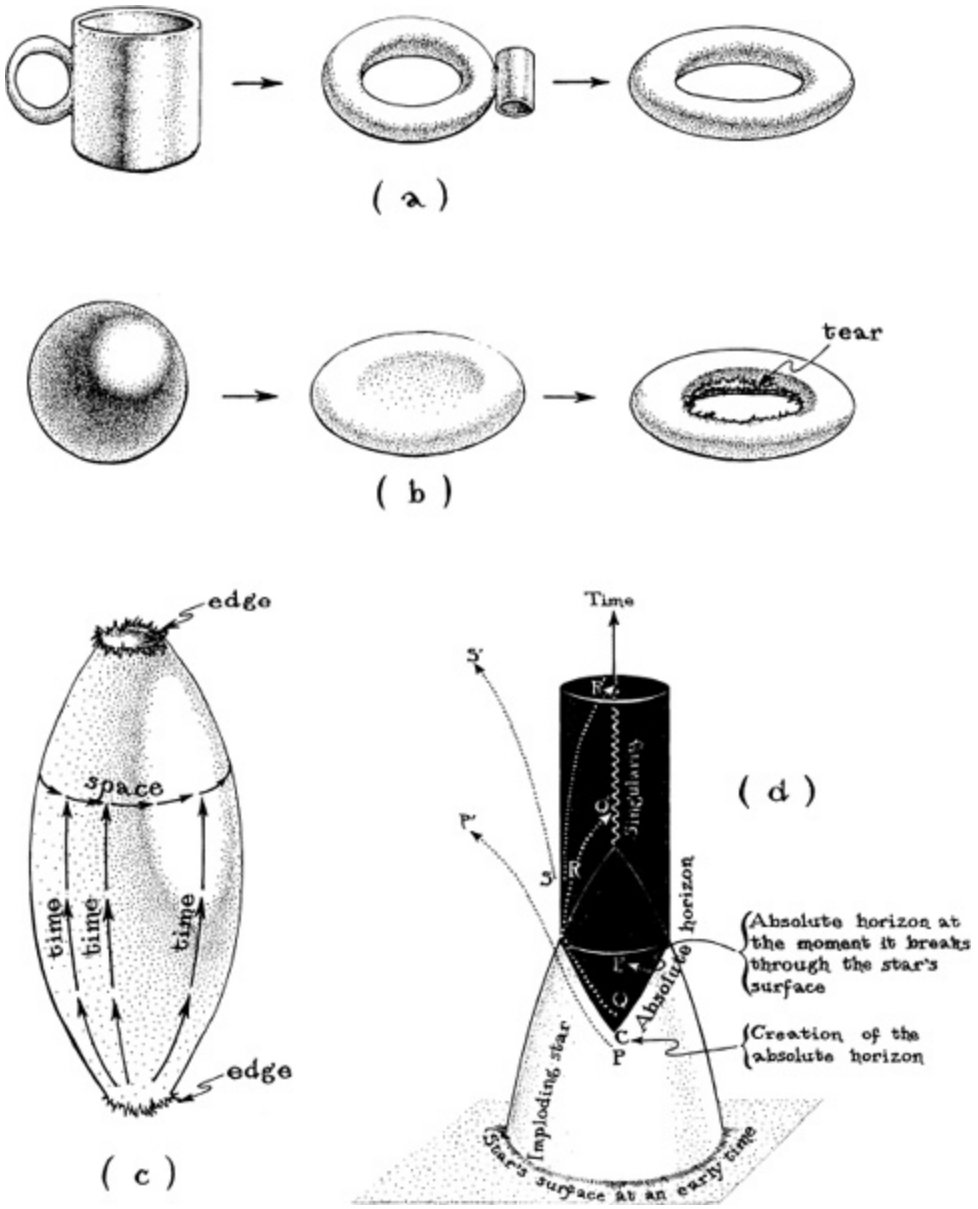


13.5 All of the following issues deal with the nature of the connections between points; that is, they are topological issues. (a) Acoffee cup (left) and a doughnut (right) can be deformed into each other smoothly and continuously without tearing, in other words, without changing the qualitative nature of any of the connections between points. They thus have the same topology. (b) To deform a sphere (left) into a doughnut (right), one must tear a hole in it. (c) The spacetime shown here has two sharp edges [analogous to the tear in (b)]: one edge at which time begins (analogous to the big bang beginning of our Universe), and one at which time ends (analogous to the big crunch). One can also conceive of a universe that has existed for all time and will always continue to exist; such a universe's spacetime would have no edges. (d) The blackened region of spacetime is the interior of a black hole; the white region is the exterior (see Box 12.1). Points in the interior cannot send any signals to points in the exterior.